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Presenting Symptoms Are Associated with Uveal Melanoma-Related Death



Posterior uveal melanoma is the most common primary malignant intraocular tumor in adults, with fatal metastatic disease developing within 10 years of primary tumor treatment in more than 25% of patients.^{1–3} Presenting symptoms include blurred vision, photopsia, floaters, visual field defects, pain, and metamorphopsia.^{4,5} We investigated their prognostic significance and report our findings herein.

The study adhered to the tenets of the Declaration of Helsinki, and approval was obtained from the regional ethical review board in Stockholm, which also waived the informed consent as this is a retrospective chart review that did not affect the treatment or follow-up of patients and did not require new collection of sensitive data. All patients treated with plaque brachytherapy for malignant melanoma originating in the choroid or ciliary body between January 1, 1980, and December 31, 2009, were included ($n = 1449$). Retrospective data were retrieved from our treatment directory according to a previously described methodology.² At the first visit, the patient was asked to describe his or her symptoms and their duration. The ophthalmologist classified and recorded them as 1 or several of (1) blurred vision (decreased visual acuity); (2) shadow in the visual field; (3) photopsia, floaters, or both; (4) metamorphopsia; (5) ocular pain; or (6) other, including photophobia, diplopia, and reduced perception of color.

Of the 1449 included patients, 649 (45%) were women and 800 (55%) were men. The mean apical tumor thickness was 5.7 mm (standard deviation [SD], 2.9 mm), with a mean largest basal diameter of 10.9 mm (SD, 4.0 mm). Median follow-up was 17.5 years (SD, 7.1 years; Table S1, available at www.aaojournal.org).

Five hundred fifty-nine patients (39%) presented with blurred vision, 277 (16%) with a shadow in the visual field; 176 (12%) with photopsia, floaters, or both; 29 (2%) with metamorphopsia; 17 (1%) with ocular pain; and 34 (2%) with other symptoms. The mean symptom duration before diagnosis was 4 months (SD, 5 months). Four hundred patients (28%) had no symptoms. One thousand two hundred fifty-three patients (86%) reported only 1 symptom, whereas 196 patients (14%) reported 2 symptoms or more. In 245 of 391 patients (63%) with recorded retinal detachment (RD) status, an RD was observed. All 245 of the RDs (100%) were exudative.

Blurred vision correlated with male gender ($\phi = 0.11$; $P < 0.001$). Patients with blurred vision were older (64 years of age vs. 62 years of age; $P = 0.02$, Student t test), showed worse visual acuity (20/300 vs. 20/120; $P < 0.001$), and had tumors with greater apical thickness (7.8 mm vs. 7.3 mm; $P < 0.001$) closer to the macula (2.7 mm vs. 3.9 mm; $P < 0.001$). However, their tumors had similar diameters (13.0 mm vs. 12.7 mm; $P = 0.51$) and similar distances to the optic disc (3.2 mm vs. 3.8 mm; $P = 0.09$). Blurred vision did not correlate with RD ($\phi = 0.06$; $P = 0.90$).

A shadow in the visual field did not correlate with gender ($\phi = 0.01$; $P = 0.59$). Patients with a shadow were older (64 years of age vs. 62 years of age; $P = 0.04$) and had tumors with greater

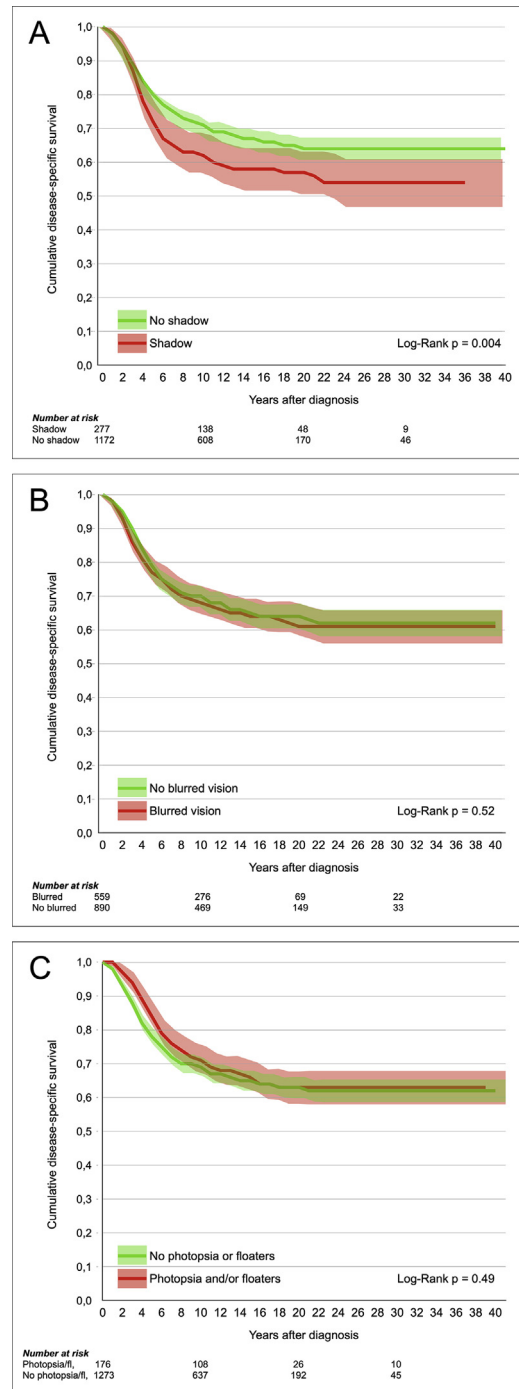


Figure 1. Graphs showing cumulative disease-specific survival proportions based on presenting symptoms. **A**, Patients with a shadow in the visual field (red) showed significantly shorter disease-specific survival than patients without a shadow (green; $P = 0.004$, log-rank test). **B**, Patients with blurred vision (red) showed similar disease-specific survival as patients without blurred vision (green; $P = 0.52$, log-rank test). **C**, Patients with photopsia, floaters, or both (red) showed similar disease-specific survival as patients without photopsia or floaters (green; $P = 0.49$, log-rank test). Colored areas represent 95% confidence intervals.

apical thickness (6.3 mm vs. 5.6 mm; $P = 0.002$) with larger diameters (11.5 mm vs. 10.8 mm; $P = 0.007$) at equal distance to the optic disc (3.6 mm vs. 3.9 mm; $P = 0.30$) and the macula (3.6 mm

vs. 3.4 mm; $P = 0.39$), but had slightly better visual acuity (20/100 vs. 20/125; $P < 0.001$). A shadow in the visual field correlated with RD ($\phi = 0.11$; $P = 0.03$).

Photopsia, floaters, or both correlated with female gender ($\phi = -0.10$; $P < 0.001$). Patients with photopsia, floaters, or both were younger (59 years vs. 62 years; $P = 0.04$) and had tumors that were more distant to the macula (4.0 mm vs. 3.3 mm; $P = 0.05$), but had similar visual acuity (20/166 vs. 20/200; $P = 0.47$) and had tumors with similar apical thickness (5.5 mm vs. 5.8 mm; $P = 0.09$), similar diameters (11.0 mm vs. 10.9 mm; $P = 0.80$), and similar distances to the optic disc (4.2 mm vs. 3.8 mm; $P = 0.19$). Photopsia, floaters, or both did not correlate with RD ($\phi = 0.06$; $P = 0.90$).

Pain did not correlate with gender ($\phi = -0.04$; $P = 0.10$). Patients with ocular pain were older (71 years of age vs. 63 years of age; $P = 0.02$) and showed worse visual acuity (20/125 vs. 20/63; $P = 0.03$), but had tumors with similar apical thickness (4.9 mm vs. 5.7 mm; $P = 0.275$), diameter (11.3 mm vs. 10.9 mm; $P = 0.69$), and distance to the optic disc (5.3 mm vs. 3.8 mm; $P = 0.36$) and the macula (5.0 mm vs. 3.4 mm; $P = 0.37$). None had extrascleral growth.

The 277 patients who presented with a shadow in the visual field had significantly shorter disease-specific survival ($P = 0.004$, log-rank test; Fig 1A). The 559 and 176 patients who demonstrated blurred vision and photopsia, floaters, or both did not have shorter disease-specific survival ($P = 0.52$ and $P = 0.49$, respectively; Fig 1B, C). Similarly, the small groups of patients with metamorphopsia ($P = 0.54$), ocular pain ($P = 0.16$), and other symptoms ($P = 0.41$) did not have shorter disease-specific survival.

In a multivariate Cox regression model with all symptoms and American Joint Committee on Cancer T category 1 through 4 as covariates, a shadow in the visual field (hazard ratio [HR], 1.5; 95% confidence interval [CI], 1.1–2.1; $P = 0.02$) and T category (HR, 2.0; 95% CI, 1.8–2.3; $P < 0.001$) were independent predictors of uveal melanoma-related mortality, but none of the other symptoms were (Table S2, available at www.aaojournal.org). A shadow in the visual field also was an independent predictor of uveal melanoma-related mortality when adjusting for tumor diameter (HR, 1.3; 95% CI, 1.0–1.6; $P = 0.03$), apical thickness (HR, 1.3; 95% CI, 1.0–1.6; $P = 0.02$), distance to the optic disc (HR, 1.5; 95% CI, 1.1–1.9; $P = 0.006$), distance to the macula (HR, 1.5; 95% CI, 1.1–1.9; $P = 0.006$), T category (HR, 1.3; 95% CI, 1.0–1.6; $P = 0.03$), and American Joint Committee on Cancer stage (HR, 1.3; 95% CI, 1.0–1.6; $P = 0.03$). Blurred vision ($P > 0.47$), photopsia, floaters, or both ($P > 0.17$), metamorphopsia ($P > 0.61$), ocular pain ($P > 0.07$), and other symptoms ($P > 0.33$) were not independent predictors of uveal melanoma-related mortality when adjusting for any one of these factors. When adjusting for all of tumor T category, stage, and distance to the macula, a shadow in the visual field retained its prognostic significance (HR, 1.3; 95% CI, 1.0–1.7; $P = 0.04$).

Limitations of this study include the difficulty in differentiating between symptoms. Some patients who reported a shadow may have experienced what others would describe as blurry vision, and so forth. Patients' descriptions of symptoms were not confirmed by additional testing, including perimetry.

In conclusion, a shadow in the visual field as a presenting symptom of uveal melanoma, experienced by 1 in 6 of 1449 patients, correlated with exudative RD and was a consistent

predictor of uveal melanoma-related mortality regardless of other simultaneous symptoms, tumor size, location, local extent, and stage. Future studies should investigate other established prognostic factors such as tumor vascularity and cytogenetic and gene expression characteristics in relationship to the experience of a shadow to reveal the causality of its association with a poor prognosis.^{6,7} Other presenting symptoms had no prognostic relevance.

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Conception and design: Stålhammar

Analysis and interpretation: Seregard, Stålhammar

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Deliberations of an International Panel of Experts on OCT Angiography Nomenclature of Neovascular Age-Related Macular Degeneration



OCT angiography (OCTA) has been used to identify and characterize macular neovascularization (MNV) secondary to age-related macular degeneration (AMD).^{1–4} Many studies have explored OCTA morphologic features of MNV that might serve as biomarkers to assess disease activity and response to treatment.^{1–6} However, the proliferation of studies has resulted in OCTA terminology that has been variable and inconsistent. To address the inconsistency of nomenclature and to allow harmonization, a multidisciplinary panel of

retinal imaging experts with a history of relevant research contributions to the field was assembled with the purpose to review published terminology and to recommend a reduced list of key terms pertinent to OCTA. The group was called UNICORN, because of its ultimate goal of generating a UNified COmmentary of the committee of inteRnational experts on the nomenclature for Neovascular AMD in OCTA. In this report, we describe the first steps of this process, which included a review of OCTA descriptors of neovascular AMD published to date and an exercise that tested agreement of these terms among retinal imaging experts.

This study was approved by the institutional review board at Tufts Medical Center and conformed to the tenets of the Declaration of Helsinki. The requirement for informed consent was waived because of the retrospective nature of the study. Before the first UNICORN meeting, a nonsystematic review of the literature was performed using the search terms *optical coherence tomography angiography* or *OCT angiography* or *OCT-A* and *neovascular macular degeneration* or *neovascular age-related macular degeneration* or *neovascular AMD* or *nAMD* or *wet age-related macular degeneration* or *wet AMD* or *wet ARMD*. A dictionary of OCTA descriptors relating to the features of MNV in AMD was generated and circulated among the panel.

At the first meeting (April 2019), UNICORN group members reviewed the list of descriptive terms on MNV resulting from AMD culled from the literature. Morphologic patterns of MNV complexes have been described by different groups with redundancies, for example, “central trunk” or “feeding vessel,” both of which are likely to have identical meanings.^{1,2,6} Patterns of vessel branching (dead tree, sea-fan, or tangle) and central and peripheral vascular arrangements within an MNV also have been described.^{1,2,6} The group identified 3 areas of concern: first, terms

Table 1. Items (Descriptors) Graded and the Intraclass Correlation Reliability Measure as Estimated Using a Mixed-Effects Logistic Model before and after Sensitivity Analysis, Respectively

Item	All Images Included				Sensitivity Analysis			
	Proportion of Scores That Are Positive (%)	Intraclass Correlation Coefficient	No. of Scores	Iterations	Proportion of Scores That Are Positive (%)	Intraclass Correlation Coefficient	No. of Scores	Iterations
Central trunk present	23.4	0.164	233	6	34.1	0.025	122	7
Lesion border well defined	53.1	0.476	234	16	73.4	0.346	121	9
Lesion resulting from AMD	83.8	0.465	245	21	82.0	0.516	127	18
MNV present	92.6	0.226	243	50	90.6	0.201	127	50
MNV hypermature	23.3	0.343	206	7	25.9	0.604	110	32
MNV immature	31.2	0.079	218	16	27.7	0.179	115	10
MNV mature	58.8	0.237	211	5	66.1	0.373	113	9
Type 1 MNV	71.3	0.531	231	43	77.2	0.235	122	50
Type 2 MNV	28.9	0.422	215	10	23.4	0.382	113	10
Type 3 MNV	6.9	0.303	217	50	1.9	0.060	113	50
PCV/aneurysmal type 1 MNV	6.7	0.550	217	34	8.3	0.859	114	47
Vessel profiles visible in central lesion	58.5	0.484	236	24	76.6	0.529	123	37
Vessel profiles visible in peripheral lesion	73.1	0.408	235	12	84.8%	0.260	122	50

AMD = age-related macular degeneration; MNV = macular neovascularization; PCV = polypoidal choroidal vasculopathy.

The mean of the individual proportions for each image's nonmissing scores that are graded positive and the number of nonmissing scores (number of scores) for each descriptor, summed across all graders that was used in the model, are shown. The number of iterations needed for convergence of the mixed-effects logistic model are presented also. When the probability of a positive score was either very high or very low, the model had difficulty converging. Model iterations to convergence were capped at 50.